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1  # Full LC Oscillator Report
2
3  ## 1. Optimization Simulation Setup
4  | Item | Value |
5  |-----|-----|
6  | VDD (V) | 1.8 |
7  | Target Frequency Fo (Hz) | 5 GHz |
8  | Frequency Offset ΔF (Hz) | 1 MHz |
9  | Temperature (°C) | 27 |
10 | Process Corner | NA |
11 | Inductor Width (μm) | 6 |
12 | Inductor Spacing (μm) | 3 |
13 | Population Size | 50 |
14 | GA Max Generations | 10 |
15 | DE Max Generations | 20 |
16 | Crossover Probability (pcross) | 0.9 |
17 | Mutation Probability (pmut) | 0.2 |
18 | Fscale | 0.7 |
19 | CR | 0.8 |
20 | Stall Generation Limit | 10 |
21 | Acceptance Tolerance | -1e9 |
22 | Stop Cost Threshold | acceptTol |
23 | PVT Corners | tt, ss, ff, sf, fs |
24 | VDD Sweep (V) | 1.62, 1.8, 1.98 |
25 | Temperature Sweep (°C) | -40, 0, 27, 70, 85 |
26
27 ---
28
29 ## 2. Best Solution Summary
30 | Parameter | Value |
31 |-----|-----|
32 | Generation | 5 |
33 | Index | 1 |
34 | Total Index | 302 |
35 | Fosc | 5.0601 GHz |
36 | Power | 3.809 mW |
37 | Vpk | 1.486 V |
38 | Vrms | 0.743 V |
39 | PN @100k | -91.09 dBc/Hz |
40 | PN @1M | -116.27 dBc/Hz |
41 | FOM @100k | -179.36 |
42 | FOM @1M | -184.54 |
43
44 ---
45
46 ## 3. Constraint Verification
47 | Metric | Target | Result | Status |
48 |-----|-----|-----|-----|
49 | Fosc | 4.9-5.1 GHz | 5.060 GHz | PASS |
50 | Power | ≤10 mW | 3.81 mW | PASS |
51 | Vpk | 0.5-1.5 V | 1.49 V | PASS |
52 | PN @1M | ≤ -100 dBc/Hz | -116.27 | PASS |
53
54 ---
55
56 ## 4. PVT Summary (Extremes)
57 - Frequency: **4.96 - 5.18 GHz**
58 - Power: **1.64 - 7.71 mW**
59 - Vpk: **1.06 - 1.93 V**
60 - PN @1M: **-119.20 to -113.20 dBc/Hz**
61 - FOM @1M: **-188.70 to -181.66**
62
63 ---
64
65 ## 5. Monte Carlo Summary
66 | Metric | Mean | Sigma |
67 |-----|-----|-----|
68 | Fosc | 5.06 GHz | 15.78 MHz |
69 | Power | 3.82 mW | 0.22 mW |

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70 | PN @1M | -116.2 | 0.86 |
71 | Vpk | 1.485 V | 0.019 V |
72 | FOM_1M | -184.5 | 0.8906 |
73
74 ---
75
76 ## 6. Yield Report
77 - Total Samples: 1000
78 - Passing Samples: 766
79 - **Yield: 76.60%**
80
81 ---
82
83 ## 7. Transient (X-Ray) Results
84 - Oscillation Frequency: **5.061 GHz**
85 - Differential Swing: **2.896 Vpp**
86 - RMS Differential: **1.033 V**
87
88 ---
89
90 ## 8. Runtime Summary
91 - Total Runtime: **2.51 hours**
92
```